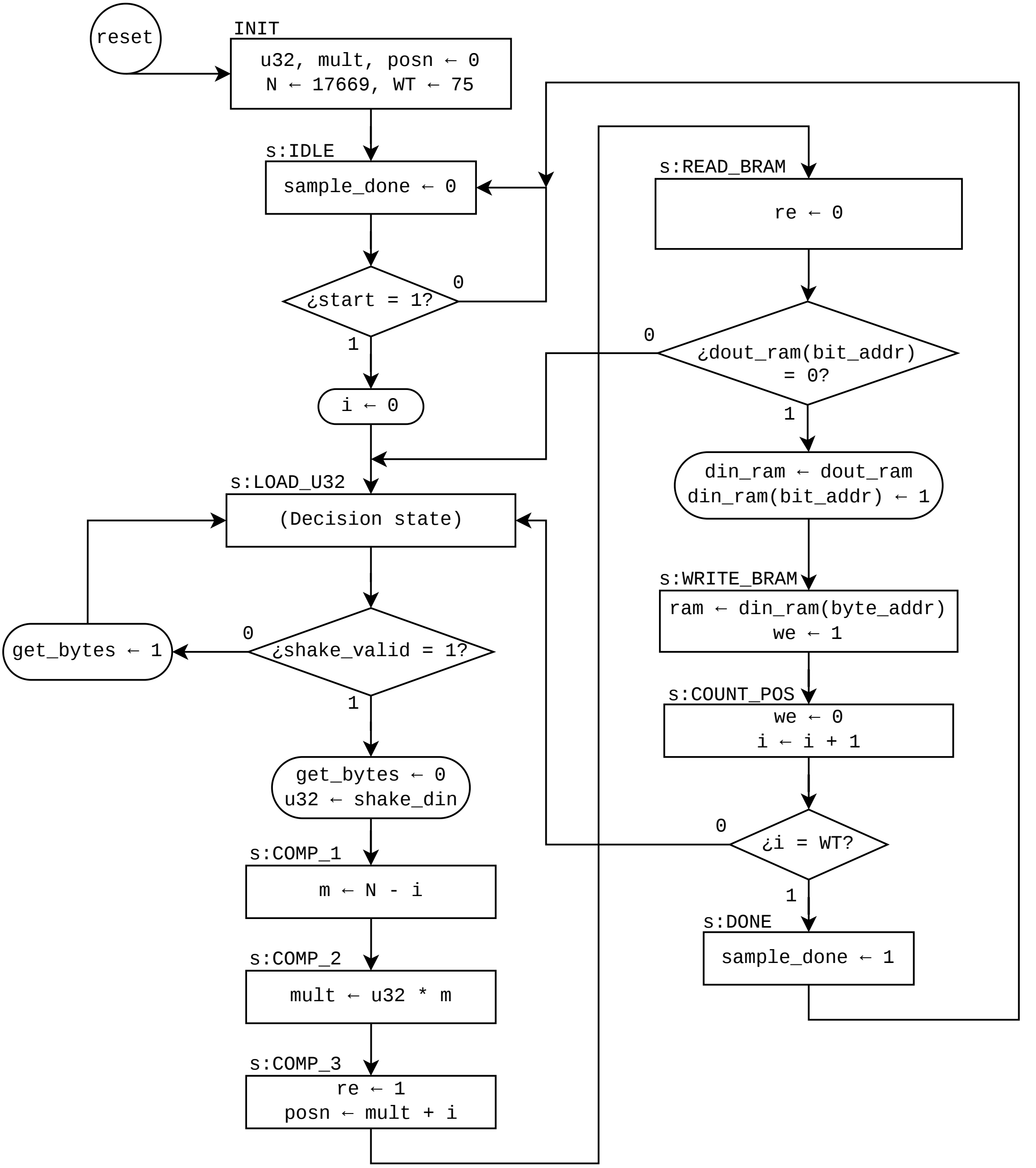
reset
INIT
u32, mult, posn ← 0
N ← 17669, WT ← 75
s:IDLE
sample_done ← 0
¿start = 1?
0
1
i ← 0
s:LOAD_U32
(Decision state)
¿shake_valid = 1?
0
1
get_bytes ← 1
get_bytes ← 0
u32 ← shake_din
s:COMP_1
m ← N - i
s:COMP_2
mult ← u32 * m
s:COMP_3
re ← 1
posn ← mult + i
s:READ_BRAM
re ← 0
¿dout_ram(bit_addr)
= 0?
0
1
din_ram ← dout_ram
din_ram(bit_addr) ← 1
s:WRITE_BRAM
ram ← din_ram(byte_addr)
we ← 1
s:COUNT_POS
we ← 0
i ← i + 1
¿i = WT?
0
1
s:DONE
sample_done ← 1